Monodromy as Memory: Quotient Commutation Is Not Quotient Injectivity

A Preprint

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Abstract

This paper gives the global theorem that Monodromy-as-Memory can carry exactly without contradicting the retained-state doctrine of Phase Calculus. For a branch surface whose loop classes are represented by a group G, and whose monodromy is a group action $\rho : G \to \mathrm{Sym}(S)$, the lifted register

$$\widehat{\Xi} = (A, q, \theta_{\text{quarters}}, \kappa, c, s, h)$$

is faithful to the retained loop class: two loop classes induce the same full-register transition for all lifted states if and only if the two loop classes are equal in the retained group. The sheet quotient satisfies the exact commutation law

$$\Pi_{\text{sheet}} \circ E_g = \rho(g) \circ \Pi_{\text{sheet}}.$$

The paper also proves the necessary negative theorem: stripped visible projection is not faithful to the full register. The double cover $z \mapsto z^2$ over the punctured plane supplies the minimal counterexample to the stronger visible-iff-full statement: the empty loop and a two-turn loop have the same stripped visible projection but different retained history and completed-turn memory. Therefore the closed global theorem is full-register faithfulness plus exact quotient commutation, not the false claim that stripped visible projection is equivalent to full lifted-state transition. The companion CLI accepts finite presentation data, branch generators, relators, words, and commutator requests, then returns sheet transitions, history words, relator checks, commutators, and projection-loss witnesses.

Keywords Phase Calculus · monodromy · finite presentations · branched covers · quotient theorem · projection loss

1 Introduction

Motivation

The branch-history register is useful only if it preserves what scalar projection erases. The finite result proves this for finite generator tables. The global upgrade must preserve the same doctrine at the homotopy-class layer: equality in the retained loop group is the equality detected by the full register, while stripped visible projection remains a quotient and therefore loses information.

The closed theorem surface

Let G be the group of loop classes supplied by a finite presentation or by any other exact group presentation of the punctured base. Let $\rho : G \to \mathrm{Sym}(S)$ be the monodromy action on finite sheets, and let $\omega : G \to \mathbb{Z}$ be the completed-turn homomorphism. The global lifted transition is

$$E_g(A, q, \theta, \kappa, c, s, h) = (A, q, \theta, \kappa + \omega(g), c, \rho(g)(s), hg).$$

This paper proves that the transition is faithful to g because the history coordinate stores right multiplication by g.

The rejected stronger statement

The statement

$$\text{same stripped visible projection} \iff \text{same full lifted-register transition}$$

is false. It contradicts the purpose of Monodromy-as-Memory: stripped projection forgets (κ, s, h). The correct equivalence is

$$E_g = E_{g'} \text{ as full-register maps} \iff g = g' \text{ in } G,$$

and the correct quotient law is

$$\Pi \circ E_g = G_g \circ \Pi$$

for a declared quotient Π, not an assertion that Π is injective.

2 Definitions

Definition 1 (Presented branch register)**.** A presented branch register is a tuple

$$\mathcal{R} = (G, S, \rho, \omega)$$

where G is a group of loop classes, S is a finite sheet set, $\rho : G \to \mathrm{Sym}(S)$ is a group action, and $\omega : G \to \mathbb{Z}$ is a completed-turn homomorphism.

Definition 2 (Global lifted state)**.** The global Monodromy-as-Memory lifted state is

$$X = (A, q, \theta, \kappa, c, s, h),$$

where $(A, q, \theta, \kappa, c)$ are inherited Phase Calculus coordinates, $s \in S$, and $h \in G$ is the retained branch-history class.

Definition 3 (Lifted transition)**.** For $g \in G$, define

$$E_g(A, q, \theta, \kappa, c, s, h) = (A, q, \theta, \kappa + \omega(g), c, \rho(g)(s), hg).$$

Definition 4 (Quotient projections)**.** The sheet projection is $\Pi_{\mathrm{sheet}}(X) = s$. The stripped visible projection is

$$\Pi_{\mathrm{vis}}(A, q, \theta, \kappa, c, s, h) = (A, q, \theta \bmod 2\pi, c).$$

3 Global Theorems M1–M5

Theorem 1 (M1: homotopy-class update closure)**.** *For every $g \in G$, E_g maps lifted states to lifted states. If $g, g' \in G$, then*

$$E_{gg'} = E_{g'} \circ E_g$$

under the convention that gg' means first traverse g, then traverse g'.

Proof. The coordinates A, q, θ, c are preserved. The coordinate $\kappa + \omega(g)$ is an integer, $\rho(g)(s) \in S$, and $hg \in G$. Thus $E_g X$ is a lifted state. For composition,

$$E_{g'}(E_g(A, q, \theta, \kappa, c, s, h)) = (A, q, \theta, \kappa + \omega(g) + \omega(g'), c, \rho(g')\rho(g)(s), (hg)g').$$

Since $\omega(gg') = \omega(g) + \omega(g')$, $\rho(gg') = \rho(g')\rho(g)$ under the traversal convention, and $(hg)g' = h(gg')$, this is $E_{gg'}X$. □

Theorem 2 (M2: exact sheet quotient)**.** *For every $g \in G$,*

$$\Pi_{\mathrm{sheet}} \circ E_g = \rho(g) \circ \Pi_{\mathrm{sheet}}.$$

Proof. For $X = (A, q, \theta, \kappa, c, s, h)$,

$$\Pi_{\mathrm{sheet}}(E_g X) = \Pi_{\mathrm{sheet}}(A, q, \theta, \kappa + \omega(g), c, \rho(g)(s), hg) = \rho(g)(s).$$

The right side gives $(\rho(g) \circ \Pi_{\mathrm{sheet}})(X) = \rho(g)(s)$. Hence the maps agree. □

Theorem 3 (M3: full-register faithfulness to loop class)**.** *For $g, g' \in G$, the following are equivalent:*

$$E_g = E_{g'} \text{ as maps on the full lifted register} \iff g = g' \text{ in } G.$$

Proof. If $g = g'$, substituting equal group elements into the definition of E gives $E_g = E_{g'}$. Conversely, suppose $E_g = E_{g'}$ as maps. Evaluate both maps on any state with history coordinate equal to the identity $e \in G$. Equality of the history coordinate gives $eg = eg'$. Since $eg = g$ and $eg' = g'$, it follows that $g = g'$. ☐

Theorem 4 (M4: stripped visible projection is not faithful)**.** *If G has a nonidentity element g, then there are lifted states X and Y such that*

$$\Pi_{\text{vis}}(X) = \Pi_{\text{vis}}(Y), \qquad X \neq Y.$$

Proof. Let $Y = E_g X$. The stripped projection keeps $(A, q, \theta \bmod 2\pi, c)$, and these coordinates are not changed by E_g. Hence $\Pi_{\text{vis}}(X) = \Pi_{\text{vis}}(Y)$. Since $g \neq e$, the retained history coordinate changes from h to hg, so $X \neq Y$. ☐

Theorem 5 (M5: quotient commutation is not quotient injectivity)**.** *The exact commutation law*

$$\Pi \circ E_g = G_g \circ \Pi$$

for a projection Π does not imply that Π is state-complete. In particular, Π_{sheet} satisfies exact quotient commutation by M2, while Π_{vis} is non-injective by M4.

Proof. M2 proves an exact quotient law for Π_{sheet}. M4 proves that Π_{vis} collapses distinct full lifted states to the same visible value. A map that collapses distinct states is non-injective. Therefore exact quotient commutation and state-completeness are different properties. ☐

4 Minimal Counterexample to the Visible-Iff-Full Claim

Proposition 1 (Double-cover counterexample)**.** *For the cover $z \mapsto z^2$ over $\mathbb{C}^*$, let $G = \pi_1(\mathbb{C}^*) \cong \mathbb{Z}$ with generator g, and let $\rho(g) = (0\,1)$. The empty loop e and the two-turn loop g^2 have the same stripped visible projection but different full lifted-register transitions.*

Proof. The monodromy action satisfies $\rho(g^2) = \rho(g)^2 = \text{id}$, so the two-turn loop returns to the same sheet. The stripped visible projection forgets κ, sheet history, and the retained word, so the empty loop and g^2 have the same stripped visible endpoint. But the full register changes history from h to hg^2 and changes completed-turn memory by $\omega(g^2) = 2\omega(g)$. Since $g^2 \neq e$ in $\mathbb{Z}$, the full register transition is different. Thus the visible-iff-full statement is false. ☐

5 Certifier

The v0.2 certifier accepts a JSON object containing S, a finite generator list, optional relators, generator permutations, winding contributions, target words, and commutator pairs. It returns relator representation checks, sheet transitions, retained history words, completed-turn deltas, commutators, projection-loss witnesses, and the double-cover counterexample when the input is the two-sheet generator $g = (0\,1)$.

The certifier evaluates finite words exactly through the supplied finite permutation representation and exact syntactic history. It does not claim to solve the word problem for arbitrary finite presentations; full group-class equality is a mathematical theorem surface and requires either the group object, a normal-form oracle, or a decidable presentation class.

6 Gate Inventory

Gate	Quantity	Pass condition	Failure meaning
G1	relators	every relator maps to identity and zero winding	representation is not a valid monodromy
G2	transition	computed end sheet equals word action on start sheet	sheet action broken
G3	history	retained word equals requested word	memory register broken

G4	commutator	noncommuting witness reports nonidentity	commutator ledger broken
G5	projection loss	same stripped projection and changed lifted state	projection-loss theorem broken
G6	counterexample	double cover falsifies visible-iff-full	false theorem not detected

7 Closure Certificate

Publication ID	PC-MM-v0.2-corrected-global-closure
Status	PROVEN for corrected global theorem
Closed theorem statements	M1–M5 plus double-cover counterexample
False requested statement	stripped visible projection iff full transition
Practical utility	extended `pc_monodromy_certifier`
Unresolved burden inside corrected scope	0
Falsified burden inside corrected scope	0

8 Conclusion

The global Monodromy-as-Memory theorem is closed at the correct retained-state boundary. Full-register transitions are faithful to retained loop classes. Sheet readout satisfies the exact quotient law. Stripped visible projection remains non-faithful, as required by the Phase Calculus doctrine that computation stays in the lift and projection happens only at the boundary. The double-cover counterexample blocks the stronger visible-iff-full claim and prevents a false PROVEN stamp from entering the publication chain.

References

No external references are required for the closed theorem surface of this bundle.